# Cloud Computing Case Study - Week 1

**Topic: Dropbox - “Magic Pocket”: Repatriation from AWS S3 to Custom Infrastructure**

Course: IT457 | Date: 20 Oct 2026

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## 1. One-Page Summary of Sources Found

The table below lists the key primary (first-party) and supporting secondary sources located for the Dropbox Magic Pocket case study. Primary sources — the Dropbox engineering blog series, Dropbox’s S-1 filing with the SEC, and conference talks by the engineering team — supply the architecture, cost, and outcome data used in the salient points.

| | Source | Type | Date | Key information extracted |
|---|---|---|---|---|
| 1 | Dropbox Tech Blog — “Magic Pocket: Scaling to Exabytes and Beyond” | Primary | Mar 2016 | Official announcement of Magic Pocket; migration of 90% of user data off AWS S3 completed on 7 Oct 2015 (almost a month ahead of the 30 Oct target), with no data loss or major disruption. |
| 2 | Dropbox Tech Blog — “Inside the Magic Pocket” | Primary | May 2016 | Detailed architecture: immutable ≤4 MB blocks, 1 GB buckets, volumes, cells (~50–100 PB), 3 geographic zones, sharded MySQL Block Index & Replication Table, Master/Volume Manager, OSD storage nodes, cross-zone async replication. |
| 3 | Dropbox S-1 Registration Statement (SEC filing, Feb 2018) - as reported by GeekWire, “Dropbox saved almost $75 million over two years…” | Primary | Feb 2018 | Cost/outcome data: cost of revenue fell $39.5M in 2016 (−$92.5M spend on third-party datacenter provider, +$53M own-datacenter expense); further $35.1M savings in 2017 → $74.6M total over two years. |
| 4 | QCon SF 2022 talk — Facundo Agriel, “Magic Pocket: Dropbox’s Exabyte-Scale Blob Storage System” (also covered by InfoQ) | Primary | Oct 2022 | Current-scale operational stats: 600,000+ storage drives, 2 PB+ per OSD using SMR drives, tens of millions of requests/sec, 99.99% availability, 12 nines durability, cold-storage tier (XOR scheme inspired by Facebook’s f4). |
| 5 | Software Engineering Daily podcast — James Cowling, “Dropbox’s Magic Pocket” | Primary | May 2016 | Design rationale and migration narrative from the storage team lead: ~3 years of work, emphasis on simplicity, control, and improved economics. |
| 6 | TechCrunch - “Dropbox begins shift to high-efficiency SMR disks” | Secon dar y | Oct 2020 | Adoption of Shingled Magnetic Recording drives to raise density and cut cost per TB. |
| 7 | InfoQ - “Magic Pocket: Dropbox’s Exabyte-Scale Blob Storage System” (article & presentation summary) | Secon dar y | 2022–2 3 | Independent technical recap; confirms 12 nines durability, 3-region North America deployment, 100+ drives per storage device, erasure coding to cut replication cost. |
| 8 | BMC / TRG / HostDime industry analyses of the Dropbox S-1 | Secon dar y | 2018 | Corroborate savings figures; note Dropbox retained ~10% hybrid AWS usage (e.g., Europe) for geographic reach and compliance. |

## 2. Case Study - Salient Points

**Background & Problem**

• Dropbox launched in 2007 storing all file content in AWS S3 while metadata ran in its own datacenters. • By 2015, with 500M+ users and hundreds of petabytes, Dropbox was one of the largest single S3 workloads. Its workload —

enormous, mostly “cold” but latency-sensitive block storage — made S3’s pay-per-use economics increasingly poor at scale.

- Key drivers for leaving the cloud: (1) raw storage unit cost, and (2) performance/latency control at extreme scale, plus the financial pressure of an approaching IPO.

**Solution Architecture (Magic Pocket)**

- A custom, self-hosted, software-defined exabyte-scale blob storage system that replaced S3 as the store for file content; metadata (FileJournal) remained on Dropbox’s own MySQL-based stack.
- Data model: files split into immutable, encrypted, ≤4 MB blocks keyed by SHA-256 hash → aggregated into 1 GB logical **buckets** → **volumes** (replicated or erasure-coded) on Object Storage Devices (OSDs).
- Cells are self-contained storage clusters of ~50–100 PB, striped across racks; three geographic zones in North America with asynchronous cross-zone replication (<1 s).
- Indexing via a sharded MySQL **Block Index** (hash → cell, bucket, checksum) and an in-memory MySQL **Replication Table** (bucket → volume → OSDs); a per-cell **Master** coordinates repairs and background work (soft-state, off the data path).
- Custom high-density OSD servers (100+ HDDs each, 1–2+ PB per machine; 8 PB+ per rack), later migrated to Shingled Magnetic Recording (SMR) drives; SSDs reserved for databases/caches.
- Design targets: 12 nines (99.9999999999%) durability, 99.99% availability; erasure coding plus a cold-storage XOR scheme (inspired by Facebook’s f4) reduce replication overhead.

**Cost & Outcome Data (from Dropbox’s 2018 S-1)**

- 2015 → 2016: cost of revenue reduced by $39.5M — third-party datacenter (AWS) spend cut by $92.5M, partially offset by +$53M for owned datacenters.
- 2017: a further $35.1M reduction in infrastructure cost → cumulative ~$74.6M operating savings over two years. • Migration executed with zero data loss and no major service disruption; 90% of data served from in-house infrastructure by 7 Oct 2015, nearly a month ahead of schedule.
- Dropbox retained a hybrid model: ~10% of workload (mainly European data-residency customers) stayed on AWS — repatriation was workload-selective, not total.
- Long-term outcome: system scaled to exabytes, 600,000+ drives, tens of millions of requests/sec, and became a textbook example of cloud repatriation done selectively at scale.

**Exercise Plan Mapping (Weeks 2–3)**

Under the “Cost & Repatriation” exercise theme, we will use cloud pricing calculators to model Dropbox’s storage workload (≈ hundreds of PB → EB class) on two providers (e.g., AWS S3 vs. Google Cloud Storage/Azure Blob), then compare against an on-premises/colocation TCO estimate (servers, HDDs, power, networking, staff) mirroring Magic Pocket, and document effort and cost differences.